University of Science and Technology

Electronic engineering

Tutorial of Physics II

COURSE COORDINATOR

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Current, Resistance, and Ohm's Law

A CURRENT (I)

$$I = \frac{q}{t}$$

amperes (1 A = 1 C/s).

A BATTERY is a source of electrical energy

the terminal potential difference of a battery is equal to its emf.

The unit for emf is the same as the unit for potential difference, the volt.

THE RESISTANCE (R)

$$R = \frac{V}{I} \qquad 1\ \Omega = 1\ \mathrm{V/A}.$$

OHM'S LAW Ohm also stated that R is a constant independent of V and I.

THE TERMINAL POTENTIAL DIFFERENCE

current I is related to its electromotive force e and its *internal resistance* r as follows:

(1) When delivering current (*on discharge*):

$$\text{Terminal voltage} = (\text{emf}) - (\text{voltage drop in internal resistance})$$

$$V = \mathscr{E} - Ir$$

(2) When receiving current (*on charge*):

$$\text{Terminal voltage} = \text{emf} + (\text{voltage drop in internal resistance})$$

$$V = \mathscr{E} + Ir$$

(3) When no current exists:

$$\text{Terminal voltage} = \text{emf of battery or generator}$$

RESISTIVITY:

$$R = \rho \frac{L}{A}$$

L in m, A in m^2, and R in Ω, the units of ρ are $\Omega \cdot m$.

RESISTANCE VARIES WITH TEMPERATURE:

$$R = R_0 + \alpha R_0 (T - T_0)$$

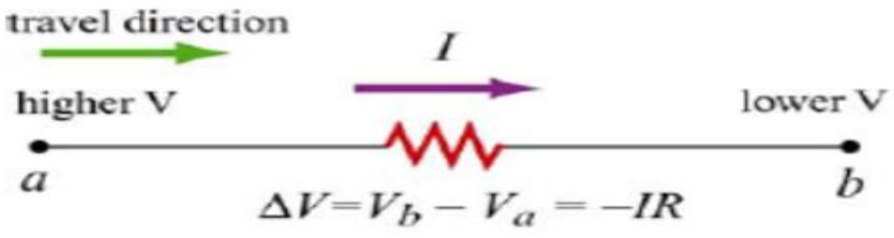
units of α K^{-1} or $°C^{-1}$.

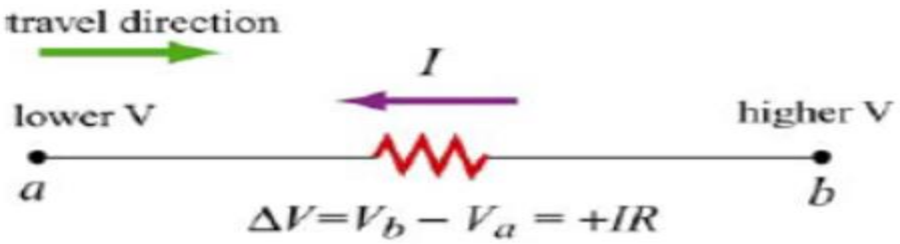
$$\rho = \rho_0 + \alpha \rho_0 (T - T_0)$$

POTENTIAL CHANGES:

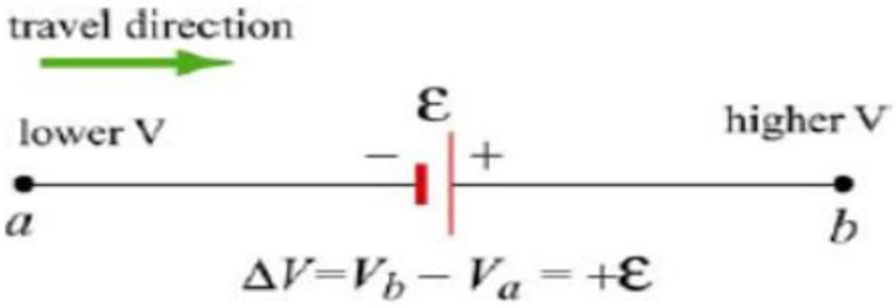
The potential difference across a resistor R

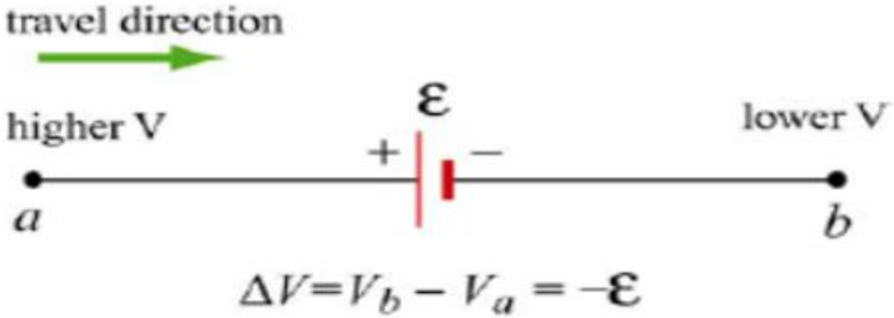
Current always flows "downhill," from high to low potential, through a resistor.

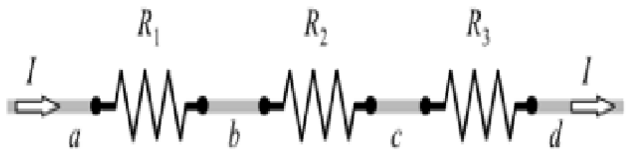




The positive terminal of a battery is always the high-potential

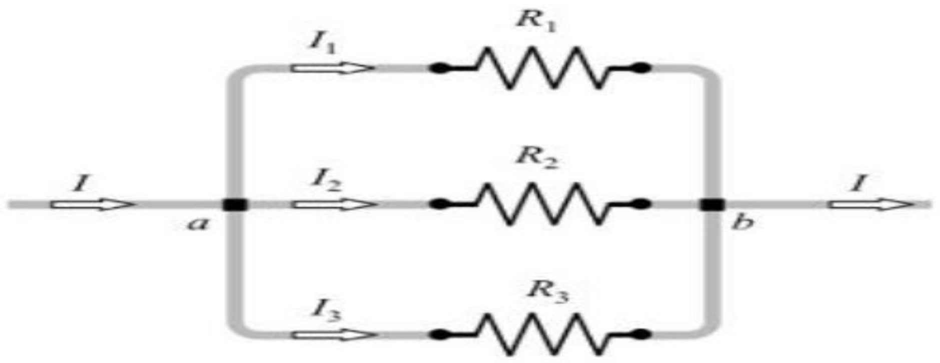






$$R_{eq} = R_1 + R_2 + R_3 + \cdots \qquad \text{(series combination)}$$

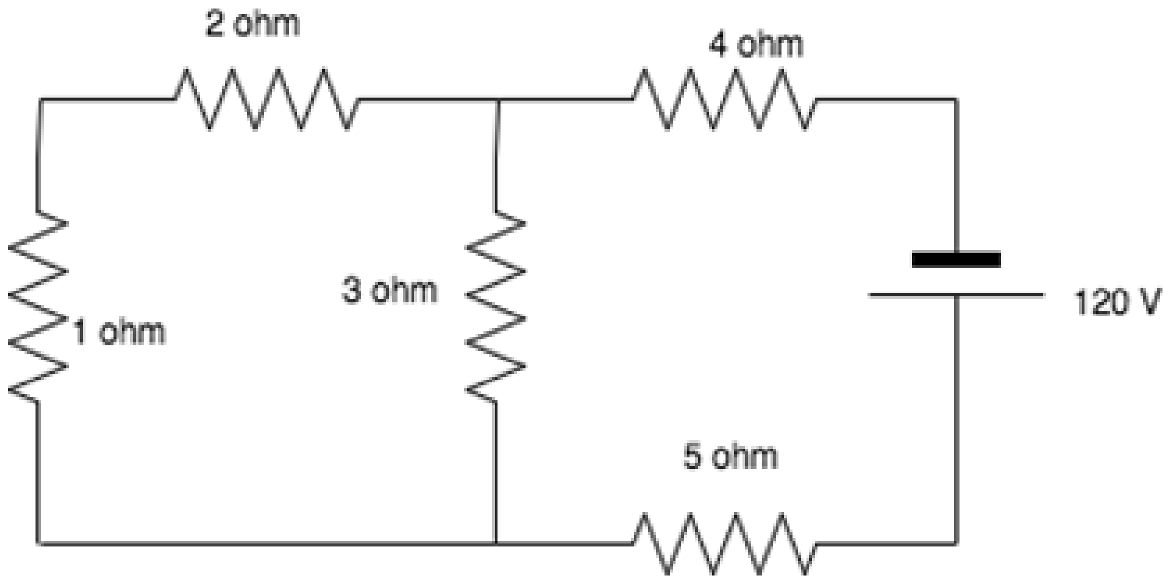
The equivalent resistance in series is always greater than the largest of the individual resistances.



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots \qquad \text{(parallel combination)}$$

The equivalent resistance in parallel is always less than the smallest of the individual resistances.

Calculate the total current in the circuit.



$$1 \ and \ 2 \ in \ sereis = 1 + 2 = 3`\Omega$$

$$3 \ and \ 3 \ in \ paraellel = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \ \therefore \ \frac{3}{2} = 1.5 \ `\Omega$$

$$1.5, 4 \ and \ 5 \ in \ series = 1.5 + 4 + 5 = 10.5`\Omega$$

$$\mathrm{I} = \frac{120}{10.5} = 11.4287\mathrm{A} = 11.43\mathrm{A}$$

How many electrons per second pass through a section of wire carrying a current of 0.70 A?

Ans. 4.4×10^{18} electrons/s

Solution:

$$q = It = 1 \times 0.7 = 0.7C$$

$$NO = \frac{q}{e} = \frac{0.7}{1.6\times10^{-19}} = 0.4375 \times 10^{-19} = 4.4 \times 10^{18} \text{ e\textbackslash s}$$

An electron gun in a TV set shoots out a beam of electrons. The beam current is 1.0×10^{-5} A. How many electrons strike the TV screen each second? How much charge strikes the screen in a minute?

Ans. 6.3×10^{13} electrons/s, -6.0×10^{-4} C/min

Solution:

$$q = It = 1 \times 10^{-5} \times 1 = 1 \times 10^{-5}C$$

$$NO = \frac{q}{e} = \frac{1 \times 10^{-5}}{16 \times 10^{-20}} = 0.0625 \times 10^{20-5} = 0.0625 \times 10^{15} = 6.3 \times 10^{13} e]s$$

$$q = It = 1 \times 10^{-5} \times 60 = 6 \times 10^{-4} C\backslash\, min$$

What is the current through an 8.0-Ω toaster when it is operating on 120 V? *Ans.* 15 A

Solution:

$$I = \frac{V}{R} = \frac{120}{8} = 15A$$

What potential difference is required to pass 3.0 A through 28 Ω? *Ans.* 84 V

Solution:

$$V = IR = 3 \times 28 = 84 \text{ V}$$

Determine the potential difference between the ends of a wire of resistance 5.0 Ω if 720 C passes through it per minute. *Ans.* 60 V

Solution:

$$I = \frac{q}{t} = \frac{720}{60} = 12A$$

$$V = IR = 12 \times 5 = 60 \text{ V}$$

Find the potential difference between points A and B in Fig. 26-5 if R is 0.70 Ω. Which point is at the higher potential? *Ans.* −5.1 V, point A

Solution:

$$(V_A - V_B) = -6 - 2 \times 3 + 9 - 3 \times 0.7 = -6 - 6 + 9 - 2.1 = -5.1\text{ V}$$

+A is high and − B is low

Compute the resistance of 180 m of silver wire having a cross-section of 0.30 mm^2. The resistivity of silver is 1.6×10^{-8} Ω·m. *Ans.* 9.6 Ω

Solution:

$$R = \rho \frac{L}{A}$$

$$L = 180\text{ m}, \qquad A = 0.3 \times 10^{-6}\text{ m}^2 \qquad \rho = 1.6 \times 10^{-8}\ \Omega\text{m}$$

$$R = \frac{1.6 \times 10^{-8} \times 180}{0.3 \times 10^{-6}} = \frac{1.6 \times 180}{0.3} \times 10^{-8+6} = 960 \times 10^{-2}\ \Omega = 9.6\ \Omega$$

A coil of wire has a resistance of 25.00 Ω at 20 °C and a resistance of 25.17 Ω at 35 °C. What is its temperature coefficient of resistance? *Ans.* $4.5 \times 10^{-4}\ °C^{-1}$

Solution:

$$R = R_0 + \alpha R_0 (T - T_0)$$

$$R = R_0\,(1 + \alpha(T - T_0))$$

$$\frac{R}{R_0} - 1 = \alpha(T - T_0)$$

$$\frac{25.17}{25} - 1 = \alpha\,(35 - 20)$$

$$0.0068 = 15\alpha$$

$$\alpha = \frac{0.0068}{15} = 4.533 \times 10^{-4}\ °C^{-1} = 4.5 \times 10^{-4}\ °C^{-1}$$

1. An electric current of 10 A is the same as ___________

a) 10 J/C

b) 10 V/C

c) 10C/sec

d) 10 W/sec

2. Consider a circuit with two unequal resistances in parallel, then ___________

a) large current flows in large resistor

b) current is same in both

c) potential difference across each is same

d) smaller resistance has smaller conductance

3. In which of the following cases is Ohm's law not applicable?

a) Electrolytes

b) Arc lamps

c) Insulators

d) Vacuum ratio values

4. Resistivity of a wire depends on ___________

a) length of wire

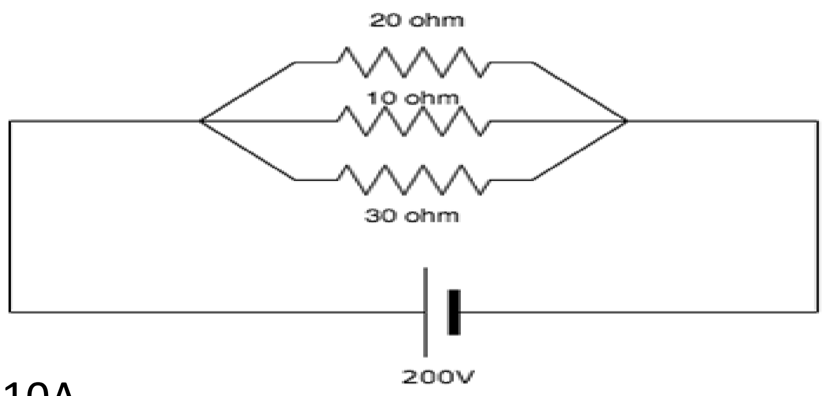
b) cross section area

c) material

d) all of the mentioned

5. Give the SI unit of resistivity.
a) ohm/metre^2
b) ohm metre^2
c) ohm meter
d) ohm/meter

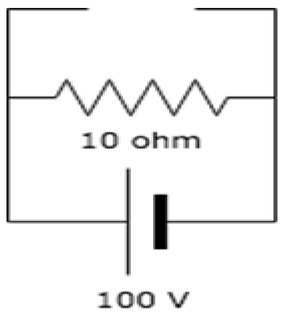
6. Calculate the current across the 20 ohm resistor.



a) 10A
b) 20A
c) 6.67A
d) 36.67A

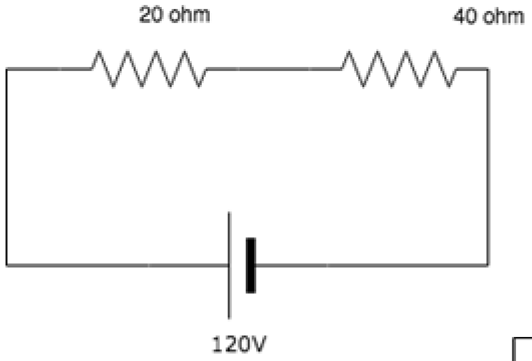
7. In a parallel circuit, with a number of resistors, the voltage across each resistor is ________
a) The same for all resistors
b) Is divided equally among all resistors
c) Is divided proportionally across all resistors
d) Is zero for all resistors

8. The voltage across the open circuit is?
a) 100V
b) Infinity
c) 90V
d) 0V



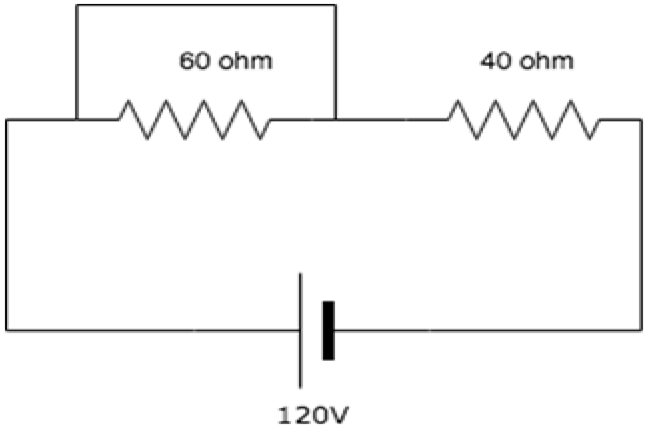
9. The currents in the three branches of a parallel circuit are 3A, 4A and 5A. What is the current leaving it?
a) 0A
b) Insufficient data provided
c) The largest one among the three values
d) 12A

10. Find the current in the circuit.



a) 1 A
b) 2 A
c) 3 A
d) 4 A

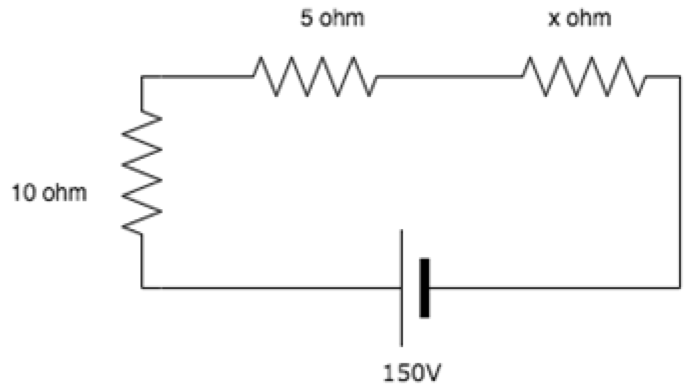
11. Voltage across the 60ohm resistor is______



a) 72V
b) 0V
c) 48V
d) 120V

12. If there are two bulbs connected in series and one blows out, what happens to the other bulb?
a) The other bulb continues to glow with the same brightness
b) The other bulb stops glowing
c) The other bulb glows with increased brightness
d) The other bulb also burns out

13. What is the value of x if the current in the circuit is 5A?
a) 15 ohm
b) 25 ohm
c) 55 ohm
d) 75 ohm



14. What is the voltage measured across a series short?
a) Infinite
b) Zero
c) The value of the source voltage
d) Null

15. It is preferable to connect bulbs in series or in parallel?
a) Series
b) Parallel
c) Both series and parallel
d) Neither series nor parallel

16. Batteries are generally connected in______
a) Series
b) Parallel
c) Either series or parallel
d) Neither series nor parallel

17. In a _________ circuit, the total resistance is greater than the largest resistance in the circuit.
a) Series
b) Parallel
c) Either series or parallel
d) Neither series nor parallel

18. In a ____________ circuit, the total resistance is smaller than the smallest resistance in the circuit.
a) Series
b) Parallel
c) Either series or parallel
d) Neither series nor parallel

19. Calculate the equivalent resistance between A and B.

a) 6.67 ohm
b) 46.67 ohm
c) 26.67 ohm
d) 10.67 ohm